

RUBRICS

S.NO	CATEGORY	DESCRIPTION	TOTAL MARKS	MARKS
1	Content	How effectively the site communicates a clear and well-defined purpose and theme, consistently maintained throughout the entire site	10	
2	Content Accuracy	All information provided by the student on the website is accurate, and all assignment requirements have been fully met	10	
3	Layout	Does the website have a beautiful and usable layout, making it easy to locate all important elements? Effective use of white space, graphic elements, and alignment is used to organize material	10	
4	Navigation	Do links for navigation are clearly labeled, and consistently placed, allow the reader to easily move from a page to related pages (forward and back), and take the reader where s/he expects to go. A Patient does not become lost.	10	
5	Fonts	The fonts are consistent and easy to read, and point size varies appropriately for headings and text. Use of font styles (italic, bold, underline) is used consistently and improves readability	10	
6	Graphics	Graphics are related to the theme/purpose of the site, are thoughtfully cropped, are of high quality, and enhance reader interest or understanding	10	
7	Images-accessibility	All images, especially those used for navigation, have an ALT tag that describes the image and its link so visually impaired people can use the Web site well.	10	
8	Spelling and Grammar	Is the final website free from errors in spelling, punctuation, and grammar?	10	
9	API Implementation	Correct implementation of Patient registration, login, profile management, item listing, and item details retrieval functionalities	10	
10	Deployment	Deployed status of the website online and on GitHub	10	
Total Marks			100	

MEDICATION-REMINDER

1. Introduction

Diabetes is a chronic condition that requires continuous management to prevent complications. For individuals with diabetes, adhering to a strict medication schedule is crucial for maintaining stable blood sugar levels. However, remembering to take medication on time can be challenging, especially for those managing multiple medications. The Medication-Reminder project is designed to address this challenge by providing an automated system that helps diabetes patients manage their medication schedules effectively. This project aims to enhance patient compliance with prescribed treatments, ultimately improving health outcomes and reducing the risk of diabetes-related complications.

2. Project Overview

The Medication-Reminder system is a digital solution that assists diabetes patients in adhering to their medication schedules. The system is designed to send timely reminders to patients, ensuring they take their medications as prescribed. The project includes the development of a mobile application that allows patients to input their medication schedules, receive reminders, and track their adherence over time. The system is customizable to accommodate different types of medications and dosing schedules, providing a personalized experience for each patient. The project also includes features such as medication logs, dosage tracking, and the ability to alert caregivers or healthcare providers if a dose is missed.

Scope:

The scope of the Medication-Reminder project includes the development and deployment of a mobile application that serves as the primary interface for patients. Key features within the scope include:

- **Medication Schedule Input:** Patients can input their prescribed medication schedules, including medication names, dosages, and times.
- **Reminder Notifications:** The system sends notifications to patients at the scheduled times, reminding them to take their medications.
- **Adherence Tracking:** The system logs when patients take their medications, allowing them to track their adherence over time.

Background:

Medication adherence is a significant challenge for diabetes patients, particularly those managing complex medication regimens. Non-adherence to medication schedules can lead to poor blood glucose control, increasing the risk of severe complications such as cardiovascular disease, neuropathy, and kidney failure. Traditional methods of medication reminders, such as pillboxes or written schedules, are often insufficient in ensuring compliance. As mobile technology becomes increasingly integrated into healthcare, digital solutions like the Medication-Reminder project offer a more reliable and Patient-friendly approach to supporting patients in managing their health.

Technologies Used:

To develop the Medication Reminder in both Android and iOS Applications, various tools, software, and system configurations are employed, with justifications as follows:

- **Programming Languages:** PHP
- **Development Tools:** Visual Studio Code.
- **Database:** XAMPP with MySQL
- **Web Framework:** None (custom PHP development)
- **Android:** Android Studio
- **iOS Application:** Xcode
- **Patient Authentication:** Firebase Authentication
- **Patient Interface Design:** Figma (for creating wireframes and visual design)

System Design:

Scenario: The Medication-Reminder project addresses the challenges faced by diabetes patients in managing their medication schedules effectively. It aims to provide a centralized platform for patients, caregivers, and healthcare providers to monitor and ensure adherence to prescribed medication regimens.

Purpose: The purpose is to optimize the medication management process for diabetes patients, making it easier for patients and caregivers to collaborate effectively. The project aims to enhance medication adherence, reduce the risk of complications, and improve overall health outcomes.

Main Features: Key features include Patient management, personalized medication schedules, automated reminders (via SMS, email, or app notifications), logging of medication intake, alerts for missed doses, and reports for healthcare providers. The system also allows for customizable alert settings and integrates with healthcare records for better tracking.

Outcomes: The Medication-Reminder project is expected to result in a well-organized and efficient system for managing medication schedules for diabetes patients. Patients will benefit from improved adherence to medication, reduced risk of health complications, and a Patient-friendly interface for managing their health.

Stakeholders: Diabetes patients, caregivers, healthcare providers (doctors, nurses), IT and support teams, pharmaceutical companies, and health insurance providers.

Project Plan

Roadmap for the Project: Provides a high-level strategy for project execution, highlighting key milestones and phases.

- Phase 1: Planning and Design
- Phase 2: Database Development
- Phase 3: iOS Development
- Phase 4: Android Development
- Phase 5: Testing and Quality Assurance
- Phase 6: Deployment and Launch

Timeline as Gantt Chart:

As the only team member, the member has full control over the project's timeline.

Task	Start Date	End Date	Duration
Project Initiation	10/07/2023	28/07/2023	18 days
Database Development	29/07/2023	26/08/2023	28 days
iOS development	27/08/2023	15/11/2023	80 days
Android development	16/11/2023	3/02/2024	79 days
Testing	4/02/2024	1/03/2024	26 days
Deployment and Launch	2/03/2024	15/03/2024	13 days

Planning and Design phase:

- Completed. This phase involved wireframing, Patient flow design, selecting design elements, and corrections made based on the reviews made.

Database Development phase:

- Completed. This phase involves the construction of database tables and queries required.

App Development phase:

- Completed. This phase involves the construction of screens of the application and connecting the backend to the front end.

Testing phase:

- Completed. This phase involves comprehensive testing and bug fixing.

Deployment:

- Scheduled for 6 weeks from now. The application will be launched on iOS and Android app.

4. Non-Functional Components and Functional Components - (Figma Prototype)**Patient Types and Their Roles:****Doctor:**

Role: The doctor is the primary administrator with overarching control.

Responsibilities:

- Add Patient, Approving Appointment, Prescribing Medicines.
- Add, update, or remove patient profiles within the App.
- Assign medication schedules tailored to each patient's needs based on their medical condition and prescription.
- Review and approve any changes to the patient's medication regimen suggested by caregivers or patients.

Patient:

Role: Adhere to their prescribed medication regimen effectively.

Responsibilities:

- Diabetes graph chart, view Medicines and Make an appointment.
- Respond to medication reminders promptly by taking the prescribed doses as indicated by the system.
- Input prescribed medications, dosages, and schedules into the system.

4. Implementation

iOS Application Development -Front-End Development

Patient Journeys (Functionalities from the Patient's Perspective):

Patient Registration:

- To successfully register a new patient in the Medication-Reminder system, the application should collect the following information: the patient's full name, residential address, mobile number, email address (optional), date of birth, gender,
- Additionally, it may include fields for any known allergies, the name of their healthcare provider, and consent to terms of service and privacy policies. All this information should be securely stored in the application's database to ensure privacy and confidentiality.

Patient Login:

- After registering, I want to log in using my Patient ID and password. The system should validate my credentials against the database, granting me access to the application.

Password Recovery:

- In the event of forgetting my password, I should be able to initiate the recovery process using my registered Patient ID. The system should then facilitate a secure password reset.

Patient Profile:

- I want to access and view my Patient profile, containing essential information such as my Patient name and personal details stored in the application's database.

Complete Assigned Tasks as Doctor:

- As a Doctor, I want to view and prescribe medicines to my patients.

Receive medication Notifications:

- I want to receive real-time notifications when doctors are prescribed to me.

Back-End Development:

Database Design:

The Medtime database consists of two tables.

1. user Table - Stores information about Patients
2. job- Stores information about the job

1. User Table:

- Patient_id (Primary Key)
- Password

2. Patient profile Table:

- Patient Id
- Patient Name
- Name
- Age
- Gender
- Phone_number

3. Patient details Table:

- Name
- Age
- Gender
- Phone_number
- Treatment

- BP
- TSH
- RBs
- Pubs
- HbA1c
- Attainder_name
- Relation of patient
- Attainder age
- Attainder gender
- Attainder phone number

4. Medication Table:

- Medication_form
- Medication name
- Morning
- Afternoon
- Night
- Before food
- After food
- First Intake
- Second intake
- Third intake

5. Testing

Testing Methodology

- Testing methodologies for the Medication-Reminder project are crucial to ensure the system's reliability, efficiency, and Patient satisfaction. Below are several testing methodologies tailored to the different aspects of the project.

Functional Testing

Unit Testing:

- Test individual components and functions (e.g., Medication notification, Patient role management) to ensure they work as intended.

Integration Testing:

- Verify that different modules and services within the system work together seamlessly (e.g., integration of the mobile app platform).

System Testing:

- Evaluate the complete and integrated application to ensure compliance with the specified requirements.

6. Deployment:

Sample Screenshots

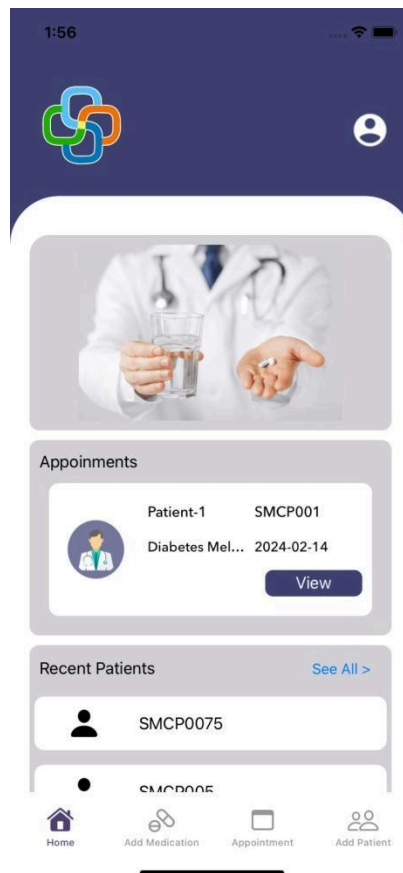


Fig 1: Home Screen

The home screen of the Medication-Reminder application prominently displays upcoming appointments and below recently added patients. When a patient ID is selected, the interface seamlessly transitions to the detailed patient profile page.

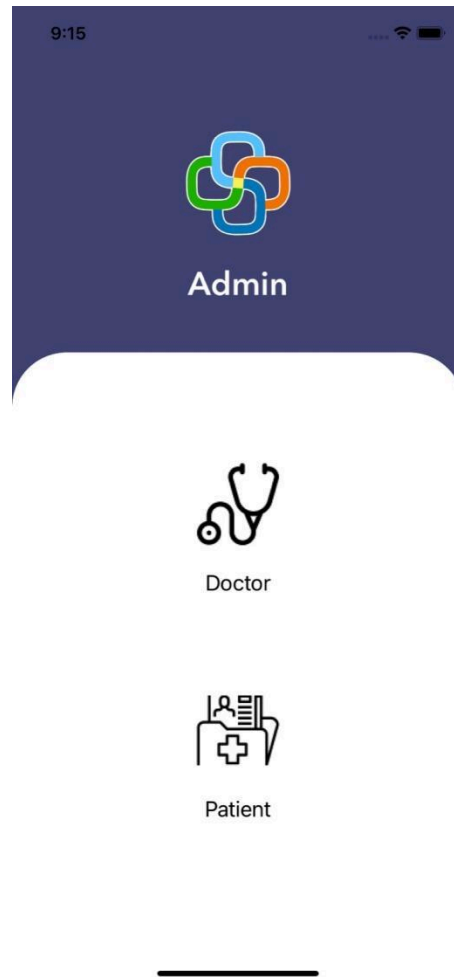


Fig 2: Admin Page

The Medication-Reminder application features two primary roles: Doctor and Patient. The Doctor's responsibilities include adding new patients, accepting appointment requests, and prescribing medications. Once the Doctor prescribes the necessary medications, the Patient receives them, and the application ensures timely notifications for the prescribed dosages.

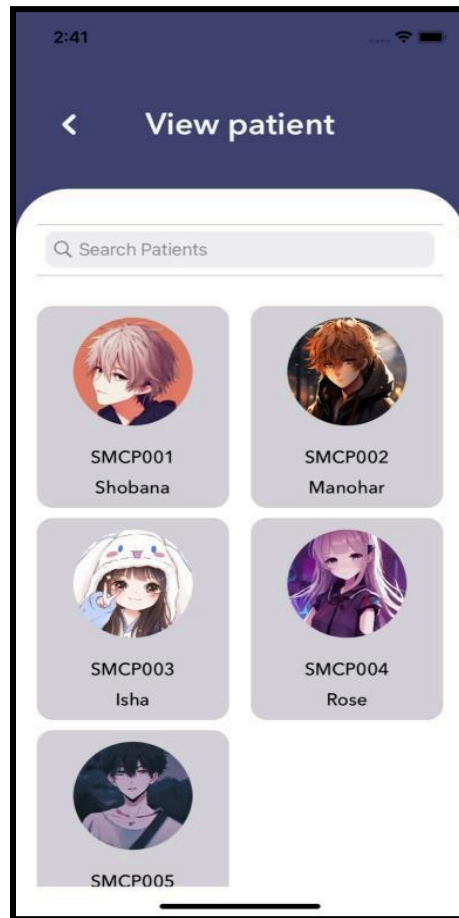


Fig 3: View the patient's Page

Displays the patient list screen of the Medication-Reminder application from the doctor login where the doctor can view all the patients. From this screen, the doctor can see the patient's name, ID, and profile. By tapping on the particular patient details he can access all the details of the patient such as personal details, points, health status, etc.



Fig 4: Medication home page

Displays the Medications screen of the Medication-Reminder application from patient login where the patient will have to view their Medications provided so that the doctor will be able to analyze the health status of the patients.

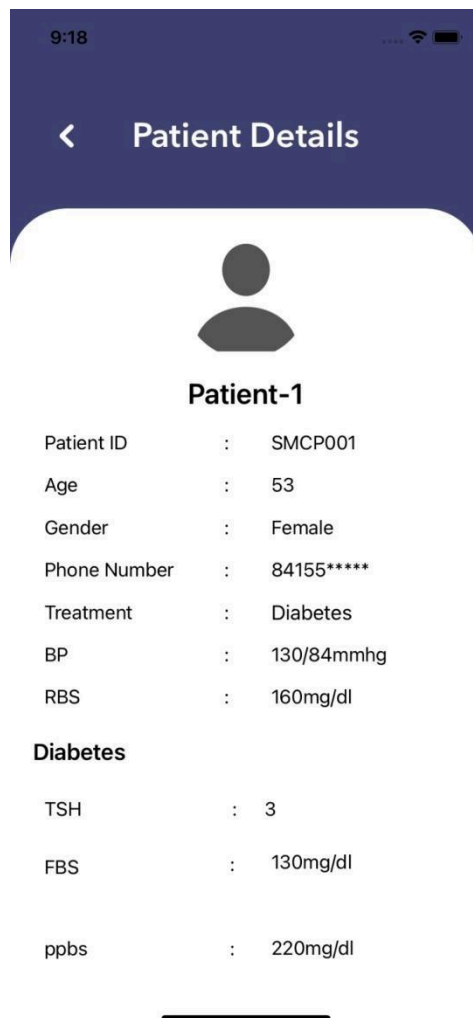


Fig 5: Patient Details Page

The figure displays the patient details of the Medication-Reminder application from the doctor login where the patient can view all medication and Attender details.

3.5 Sample Coding

dbh.php

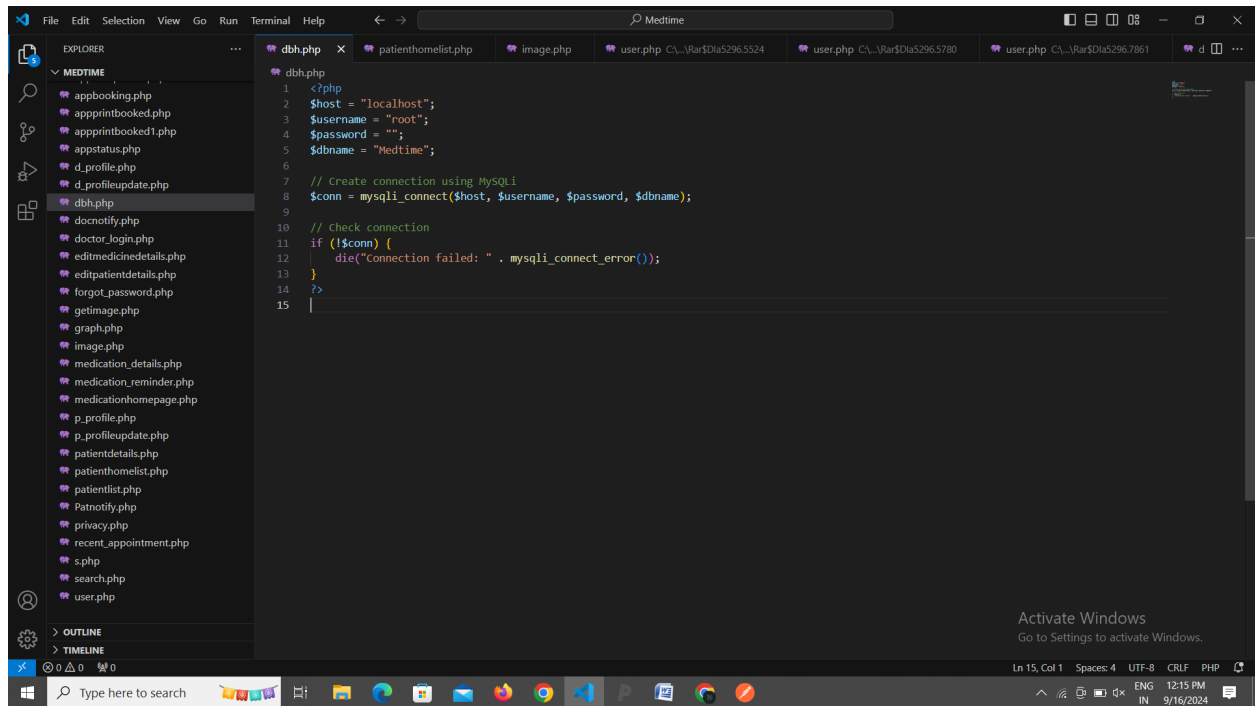


Fig 6: dbh.php typically serves as the main entry point for PHP-based both Android and iOS application

user.php

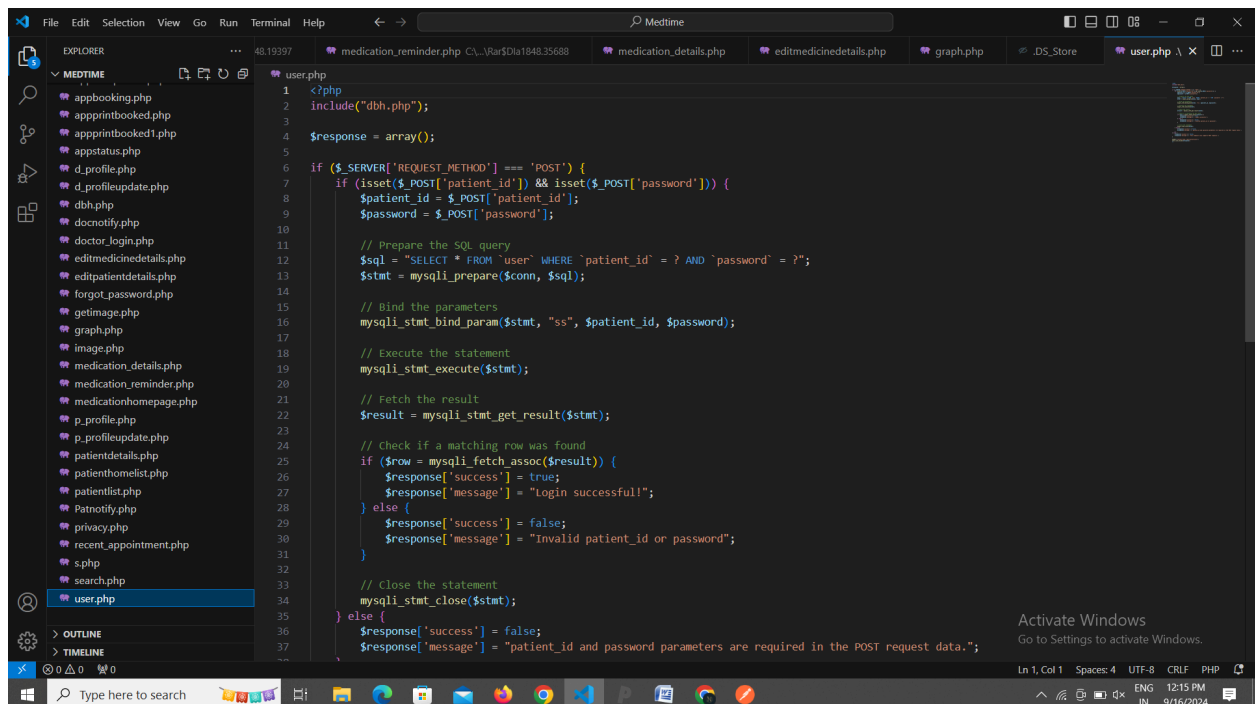


Fig 7: login.php manages Patient authentication and session initiation for secure access to both Android and iOS Applications.

Medication.php

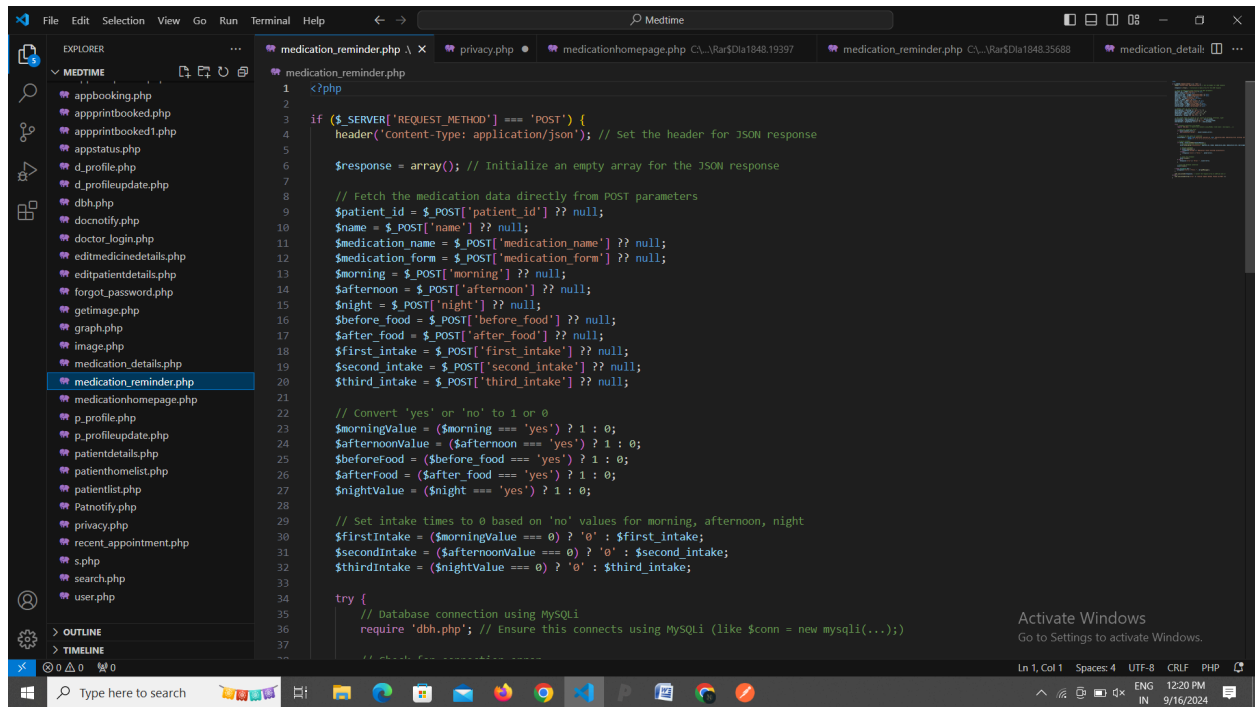


Fig 8: Medication.php typically serves patients to prescribe the medicine.

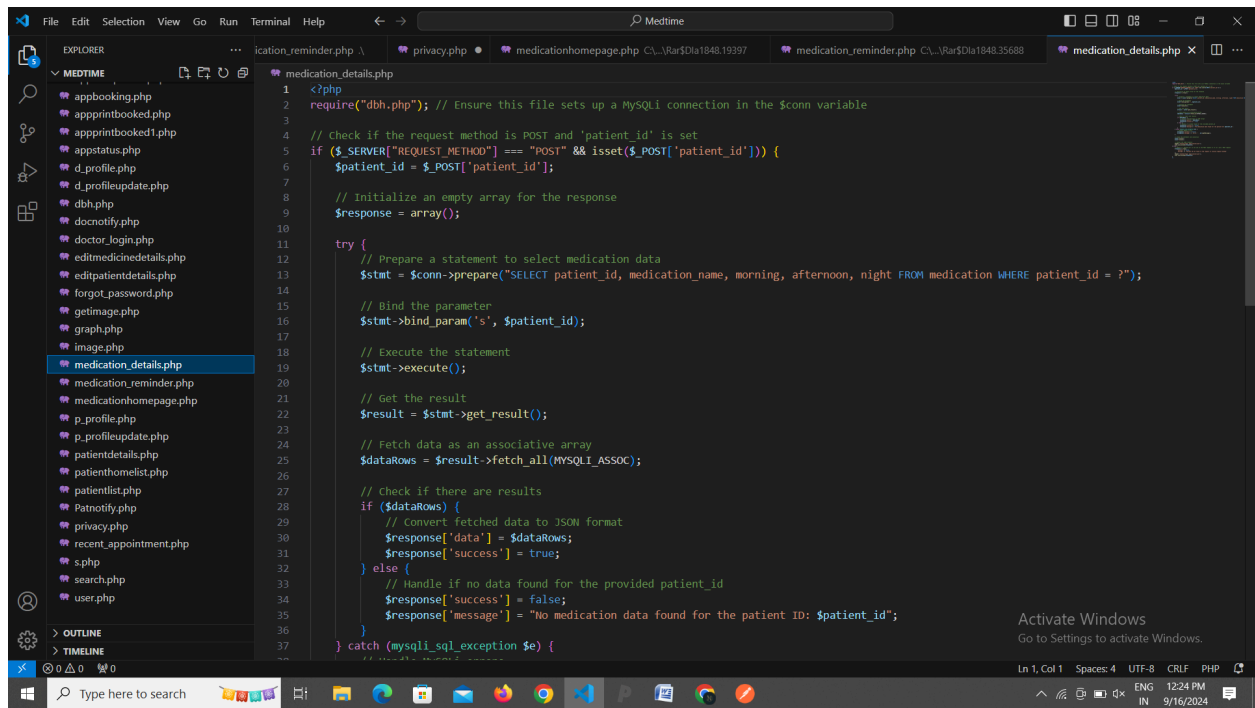


Fig 9: Medication_details.php shows details of the medication that the doctor has been prescribed.

7. Challenges and Solutions

Summary of Changes



Fig 10: Medication Home Page

Displays the Medications screen of the Medication-Reminder application from patient login where the patient will have to view their Medications provided so that the doctor will be able to analyze the health status of the patients.

8. Future enhancements:

The Housekeeping Management project can further elevate its functionality, Patient experience, and overall impact. Here are several potential enhancements:

Predictive Maintenance: Utilize AI to predict when certain areas or assets will require cleaning or maintenance, based on usage patterns and data.

Task Optimization: Employ machine learning algorithms to optimize task assignment and scheduling, ensuring the most efficient use of resources.

9. Conclusions:

The Housekeeping Management project stands as a beacon of innovation and efficiency in the realm of building maintenance. By integrating the roles of Admin, Manager, Supervisor, and Store Keeper into a cohesive and Patient-friendly application available on web platforms, the project ensures that housekeeping is not just a task but a well-coordinated effort. The emphasis on seamless communication, precise task assignment, and efficient asset management underscores the project's commitment to fostering a clean, organized, and well-maintained environment. As we embrace this sophisticated system, we anticipate a transformative impact on how buildings are maintained, marking a significant advancement in housekeeping management practices.

10. Annexure

Github link

https://github.com/nn1986325/Medication_Reminder

URL

<https://play.google.com/store/apps/details?id=com.simats.medtime>